

CHUNRU LIN

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EDUCATION

University of Massachusetts Amherst, Amherst, MA, USA

Ph.D. in Computer Science

Sept. 2023 - Present

advised by Prof. Chuang Gan

Shanghai Jiao Tong University, Shanghai, China

B.E. in Computer Science, ACM Honors Class

Sept. 2019 - June 2023

Overall GPA: 3.8/4.3 (Rank: 5/29)

PUBLICATIONS

1. **RoboWits: Unexpected Challenges for Robotic Creative Problem Solving.** Chunru Lin*, Hongxin Zhang*, Fenghao Yu, Zhehuan Chen, Thomas L. Griffiths, Yejin Choi, David Held, Chuang Gan. Under review of **NeurIPS 2026**.
2. **Deformable Linear Object Manipulations with Differentiable Physics.** Junyi Cao, Yian Wang, Ziyang Xiong, Chunru Lin, Zhehuan Chen, Yi-Ling Qiao, Minghao Guo, Dieter Fox, Chuang Gan. Accepted by **ICML 2026**.
3. **RobotSmith: Generative Robotic Tool Design for Acquisition of Complex Manipulation Skills.** Chunru Lin, Haotian Yuan, Yian Wang, Xiaowen Qiu, Tsun-Hsuan Wang, Minghao Guo, Bohan Wang, Yashraj Narang, Dieter Fox, Chuang Gan. Accepted by **NeurIPS 2025**.
4. **Lucibot: Automated robot policy learning from generated videos.** Xiaowen Qiu, Yian Wang, Jiting Cai, Zhehuan Chen, Chunru Lin, Tsun-Hsuan Wang, Chuang Gan. Preprint.
5. **TopoGaussian: Inferring Internal Topology Structures from Visual Clues.** Xiaoyu Xiong, Changyu Hu, Chunru Lin, Pingchuan Ma, Chuang Gan, Tao Du. Accepted by **ICLR 2025**.
6. **UBSoft: A Simulation Platform for Robotic Skill Learning in Unbounded Soft Environments.** Chunru Lin, Jugang Fan, Yian Wang, Zeyuan Yang, Zhehuan Chen, Lixing Fang, Tsun-Hsuan Wang, Zhou Xian, Chuang Gan. Accepted by **CoRL 2024**.
7. **3D Concept Learning and Reasoning from Multi-View Images.** Yining Hong, Chunru Lin, Yilun Du, Zhenfang Chen, Joshua B. Tenenbaum, Chuang Gan. Accepted by **CVPR 2023**.
8. **DiffVL: Scaling Up Soft Body Manipulation using Vision-Language Driven Differentiable Physics.** Zhiao Huang, Feng Chen, Yewen Pu, Chunru Lin, Hao Su and Chuang Gan. Accepted by **NeurIPS 2023**.
9. **3D Concept Grounding on Neural Fields.** Yining Hong, Yilun Du, Chunru Lin, Joshua B. Tenenbaum, Chuang Gan. Accepted by **NeurIPS 2022**.

PROJECTS

1. **Genesis: A Generative and Universal Physics Engine for Robotics and Beyond.** This is a unified physics simulation platform for robotics and embodied AI. It combines a robust physics engine, fast simulation, photo-realistic rendering, and a generative data engine. The platform is open-sourced and continuously evolving.

INTERNSHIPS

Genesis AI <i>Research Scientist Intern, working with Johnson Wang, Yiling Qiao and Zhenjia Xu</i>	Dec. 2025 - Present
Dolby Laboratory Inc. <i>PhD Research Intern, working with Vijay Sundaram and Lie Lu</i>	May. 2024 - August. 2024
Shanghai Qi Zhi Institute, China <i>Research Intern, working with Prof. Tao Du</i>	Mar. 2023 - Sept. 2023
Massachusetts Institute of Technology <i>Visiting Student Researcher, advised by Prof. Josh Tenenbaum</i>	May. 2022 - Dec. 2022

TEACHING EXPERIENCE

Teaching Assistant <i>Neural Networks: A Modern Introduction, Programming Methodology</i>	2024-2025
Teaching Assistant <i>Computer System</i>	2021-2022
Student Instructor <i>Principle and Practice of Computer Algorithms</i>	2020

DISCIPLINE COMPETITIONS

2021 CCPC Girls' Competition	Gold Medal, The Third Place
2019 CCPC, Qinhuangdao	Silver Medal, Best Women's Team
2019 CCPC Final	Silver Medal, Best Women's Team
2019 ICPC, Jakarta	Bronze Medal
2018 National Olympiad in Informatics, China	Silver Medal

*ICPC: International Collegiate Programming Contest *CCPC: Chinese Collegiate Programming Contest

SKILLS AND INTERESTS

Programming <i>Python/C++/Java/Rust/Verilog/JavaScript</i>
Language <i>Mandarin (native), English, Sanskrit (Beginner)</i>

AWARDS

ICML Gold Reviewer, Program Committee	2026
Spaulding-Smith Fellowship, UMass Amherst	2023
Academic Excellence Scholarship, SJTU (Top 5%)	2022
Zhiyuan Honorary Scholarship	2020, 2021, 2022